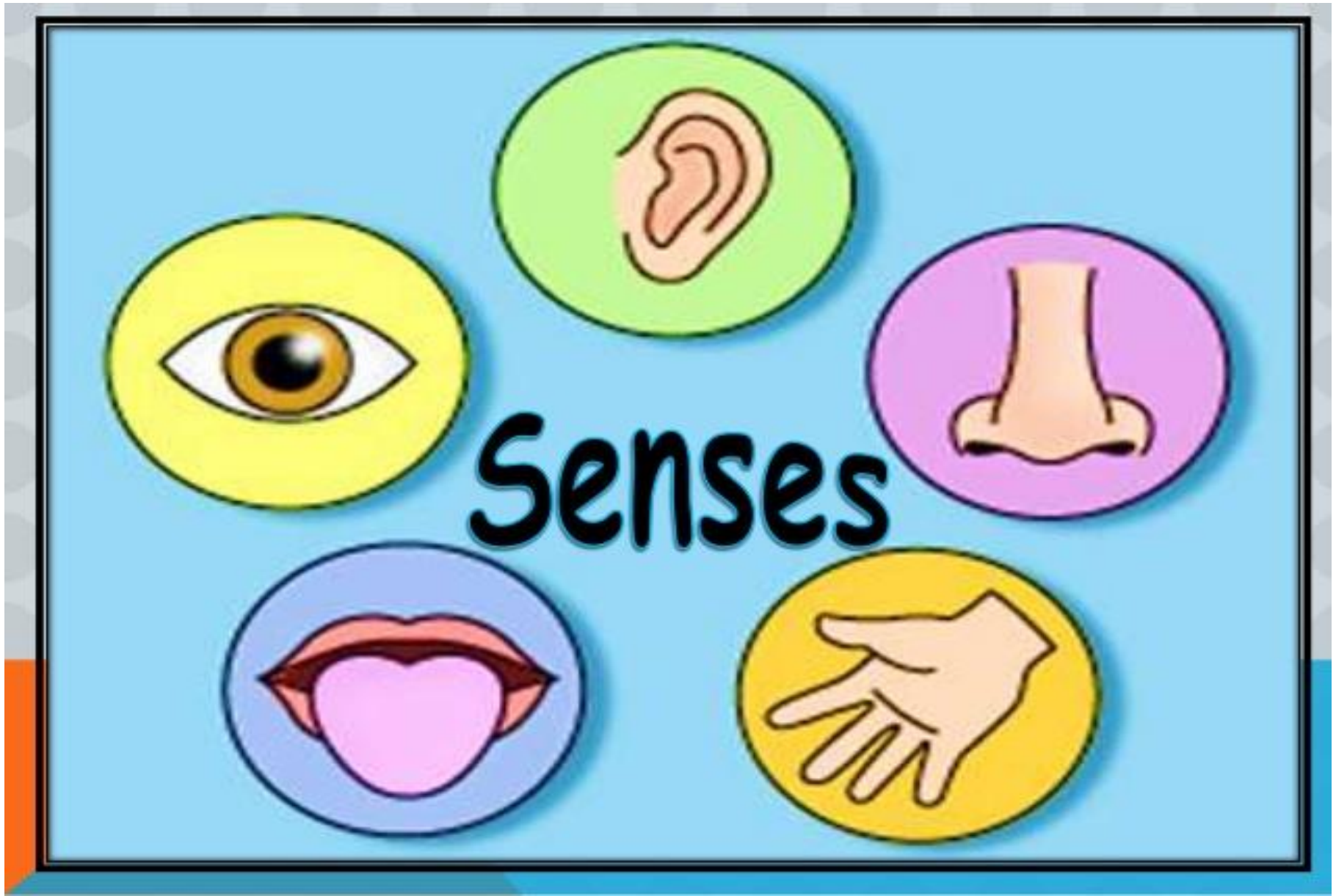


Chapter Two: Sensation & Perceptions



Chapter two :sensation and Perception

Sensation and Perception



- Ψ All outside information comes into us through our senses.
- Ψ **Sensation**—the process of detecting, receiving, converting and transmitting information resulting from stimulation of sensory receptors.
- Ψ **Perception**—the process of selecting, identifying, organizing and interpreting sensory input into a useful and meaningful mental representations of the world in the light of relevant memories from past experiences.

Chapter two :sensation and Perception

Sensation & Perception

Sensation is the impact of a stimulus on receptor cells in our sensory organs: the eyes, ears, nose, tongue, and fingertips (among other parts of the body that sense feeling/touch). Stimuli from the environment are transformed into neural signals.

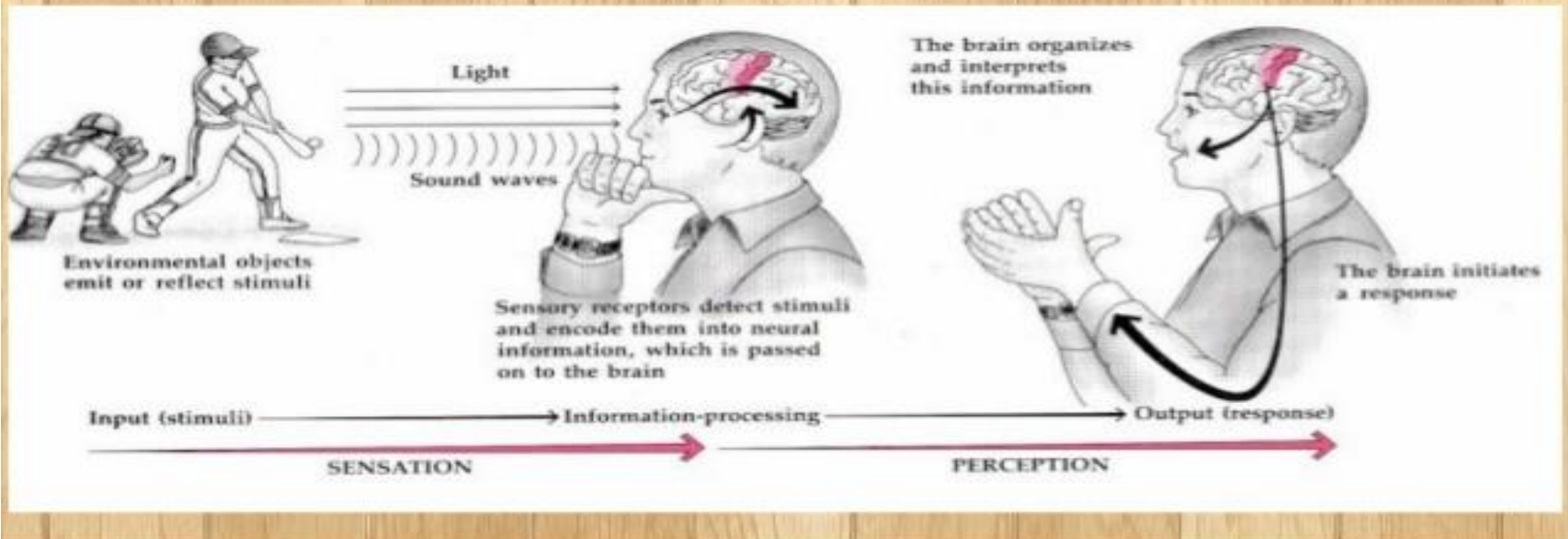


Perception is the mental process of understanding the sensory input received in our brains: it is awareness or understanding of the real-world cause of the sensory input. The neural signals we receive from our sensory organs are processed in the brain, and the brain creates useful information and meaning about the world around us.

Chapter two :sensation and Perception

Sensation & Perception

In psychology, sensation and perception are stages of processing the sensory systems, such as vision, auditory, and pain sensory systems.



Chapter two :sensation and Perception

Sensation & Perception



Sensation + Perception = our interpretation of reality around us.

Chapter two :sensation and Perception

The Processes of Sensation and Perception

Information (e.g. light, sound)—activate our **sense receptors** in the sensory organs which receive and process sensory information from environment.



Transduction—after stimuli enter sensory organs, the sense receptor will change/covert the stimulus into electrical signals called **neural impulses** which are sent to the brain.

When neural impulses reach the particular area in the brain, they are changed into meaningless bits of information called **sensation** which involves the *detection* of sensory stimuli.

These meaningless bits of information are then changed into meaningful and complete images called **perception**—the *interpretation* of sensory stimuli.

Chapter two :sensation and Perception

PERCEPTUAL PROCESSES

- Perceptual processes include:
 - **Selection** refers to choosing which of many stimuli that will be processed.
 - **Organization** involves collecting the information into some pattern.
 - **Interpretation** involves understanding the pattern.

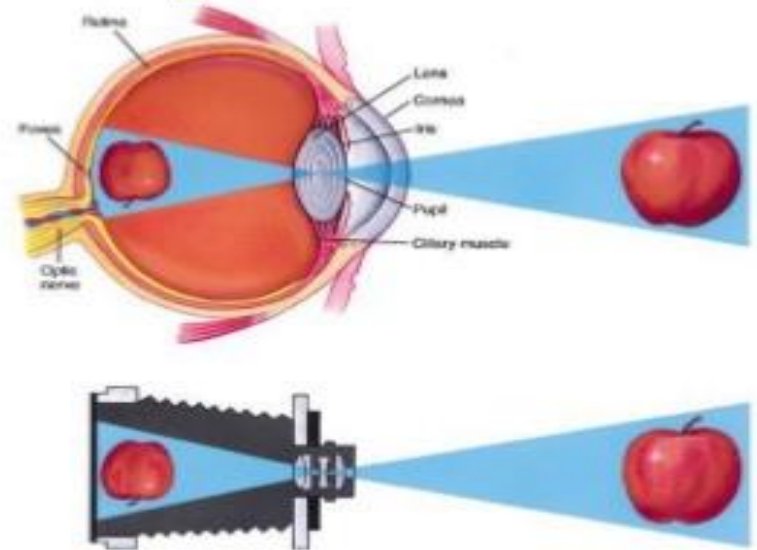
Chapter two :sensation and Perception

How Sensation and Perception Work?

Ψ Our sense organs translate physical energy from the environment into electrical impulses processed by the brain.

For example, light, in the form of electromagnetic radiation, causes receptor cells in our eyes to activate and send signals to the brain.

Ψ But we do not understand these signals as pure energy. The **process of perception** allows us to interpret them as objects, events, people, and situations.



Chapter two :sensation and Perception

Basic Function



- Ψ The basic function of sensation is **detection** of sensory stimuli, whereas perception generally involves **interpretation** of the same stimuli.
- Ψ Our senses tell us something is out there. Our perception tell us what that something is.
- Ψ In practice, sensation and perception are virtually impossible to separate, because they are part of one continuous process.

Chapter two :sensation and Perception

Transduction

Transduction is the process of turning environmental information into neural impulses.



- In the eyes, light waves create neural impulses



- In the nose, chemical reactions from gaseous molecules create neural impulses



- On the tongue, chemical reactions create neural impulses



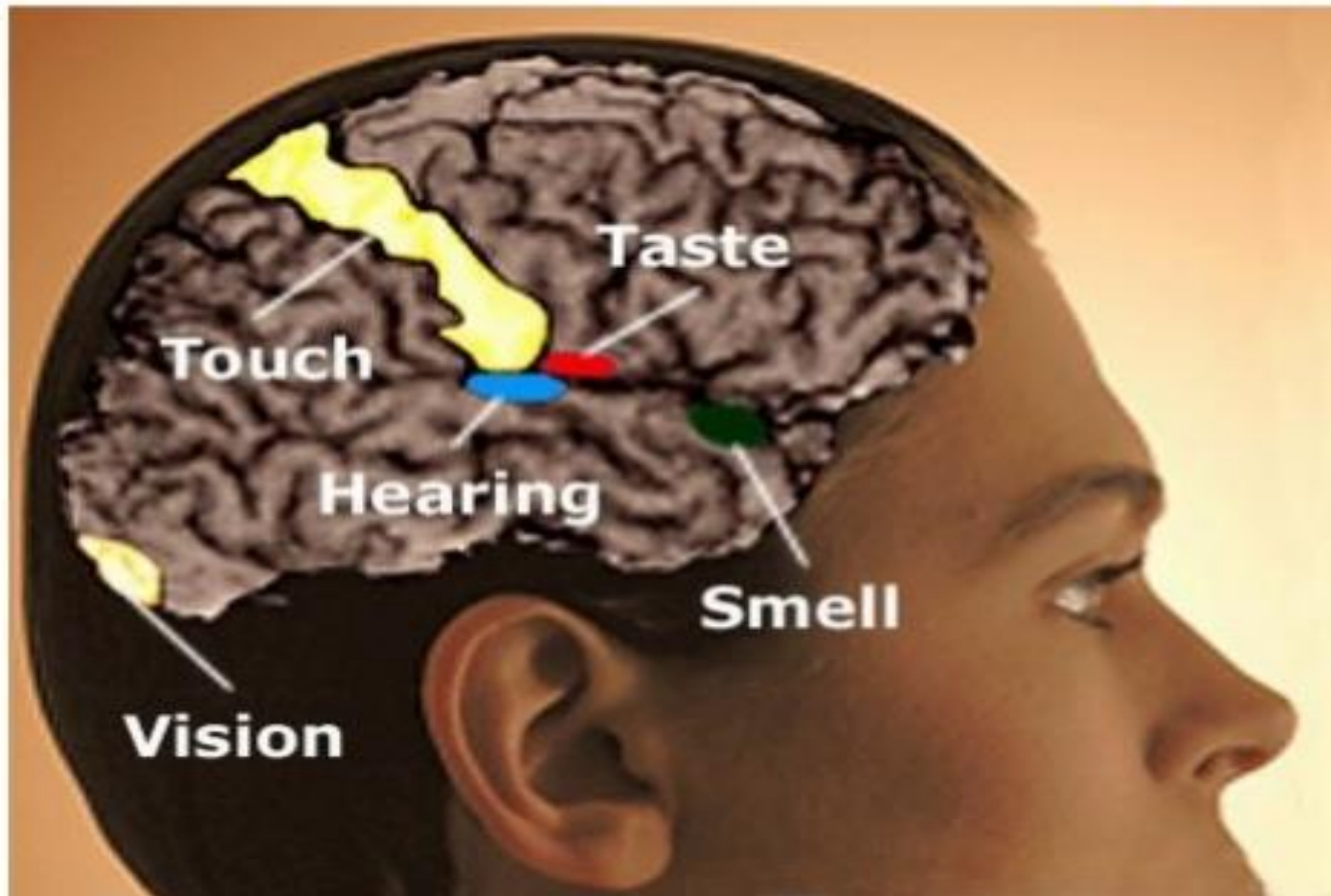
- In the ear, sound waves create neural impulses



- Within the skin, pressure, pain, and temperature create neural impulses

Chapter two :sensation and Perception

Parts of the Brain Stimulated by Senses



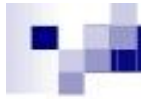
Chapter Two: sensation and Perception

Sensory

THRESHOLD



Chapter Two: sensation and Perception



Absolute Threshold

- The **Absolute Threshold** is the level of stimulus that produces a positive response of detection 50% of the time.
- the lowest level of stimulation that a person can consciously detect 50 percent of the time the stimulation is present.

Chapter two :sensation and Perception

Absolute Threshold

The weakest amount of a stimulus that a person can detect 50% of the time.

Sight		Seeing a candle flame 30 miles away on a clear night
Hearing		Hearing a watch ticking 20 feet away
Touch		Feeling a bee's wing falling a distance of 1cm onto your cheek
Smell		Smelling one drop of perfume in a three room house
Taste		Tasting one teaspoon of sugar dissolved in two gallons of water

Chapter two :sensation and Perception

Difference Threshold

The smallest amount of change in a physical stimulus that a person can detect 50% of the time. This is also called the “just noticeable difference.”

If someone turns the music up slowly, at what point do you notice it has become louder?

If you hold a handful of sand, and someone adds one grain at a time to the pile, when do you notice it has become heavier?

If your best friend trims a half inch off of their hair, will you notice the difference?

Chapter Two :laws of principles of perception

Law of Proximity

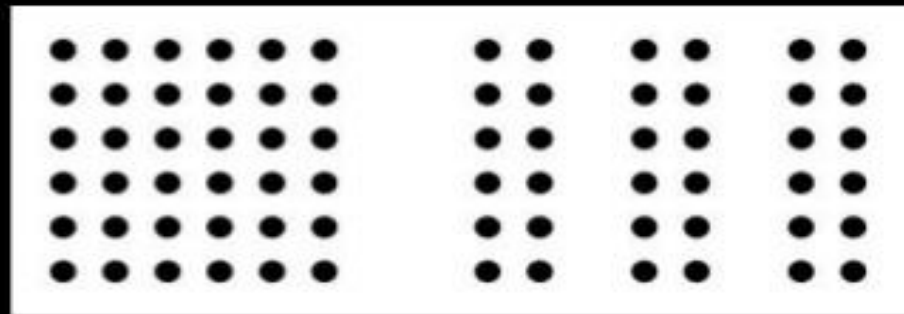
Law of Proximity —

Spatial or temporal proximity of elements may induce the mind to perceive a collective or totality.

Chapter Two : laws of perception

Law of Proximity

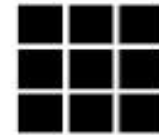
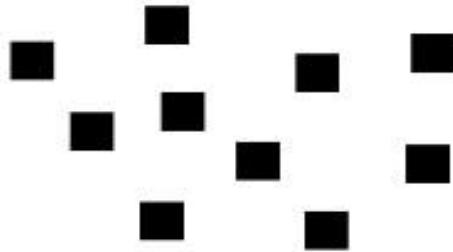
Elements that are placed close to each other will often be perceived as one group.



When the dots are placed close to each other in groups, we tend to perceive 3 columns rather than a whole group of dots.



Chapter Two : laws of perception



Law of Proximity

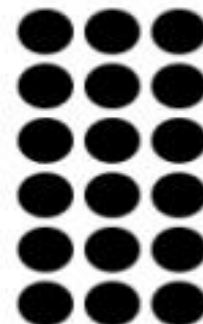
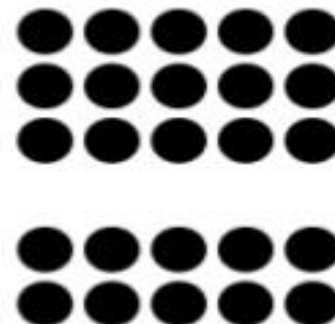
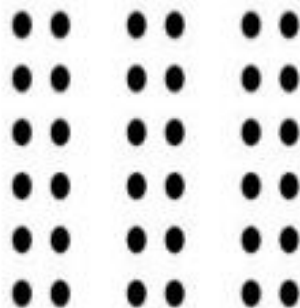
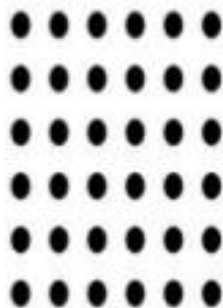
Spatial or temporal proximity of elements may induce the mind to perceive a collective or totality.

Chapter Two : laws of perception

Gestalt Principles

Law of Proximity

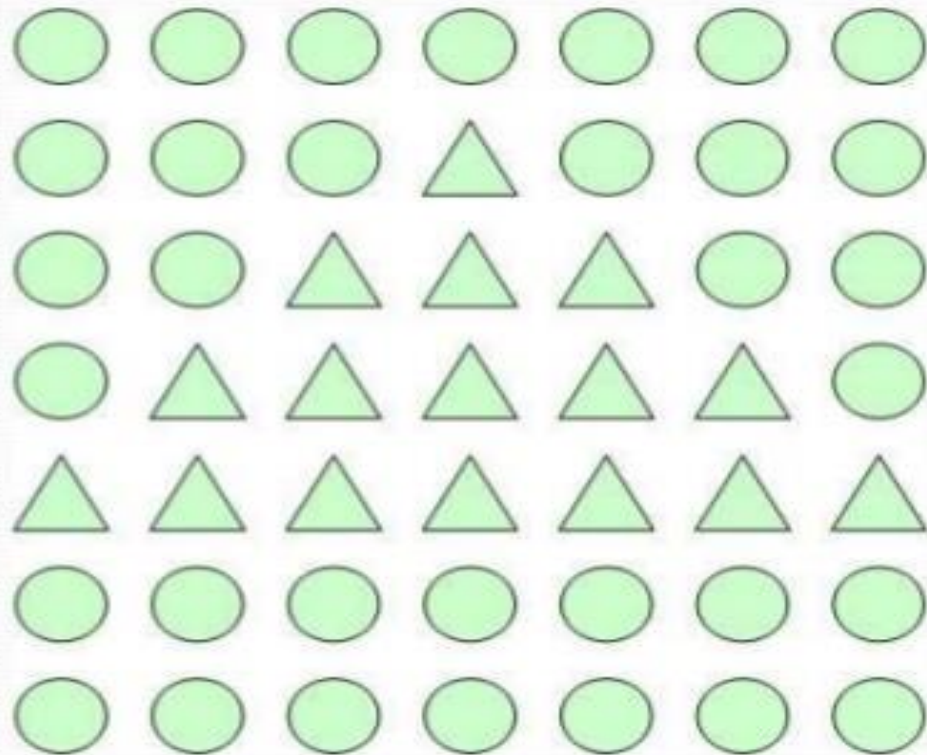
- Elements that are close together will be perceived as a coherent object.



Chapter Two : laws of perception

Law of Similarity

Elements that look similar will be perceived as part of the same form. There seems to be a triangle in the square. We link similar elements together



Chapter Two : laws of perception

Law of Similarity



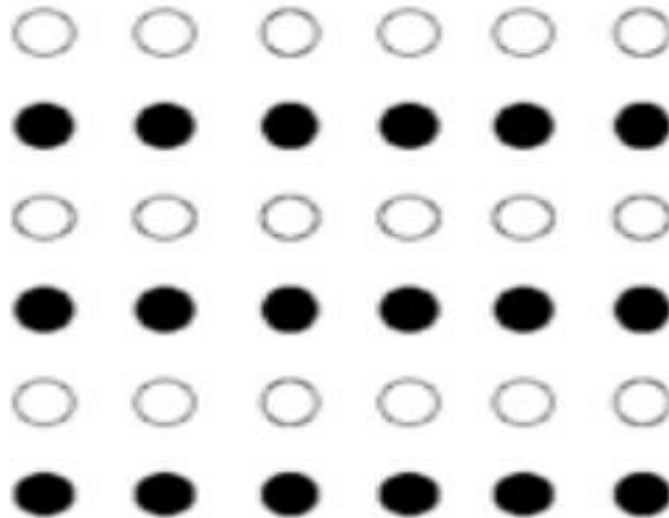
Law of Similarity:

Items that are similar tend to be grouped together.

In the image above, most people see vertical columns of circles and squares.

Chapter Two : laws of perception

Law of Similarity



Chapter Two : laws of perception

Law of Similarity

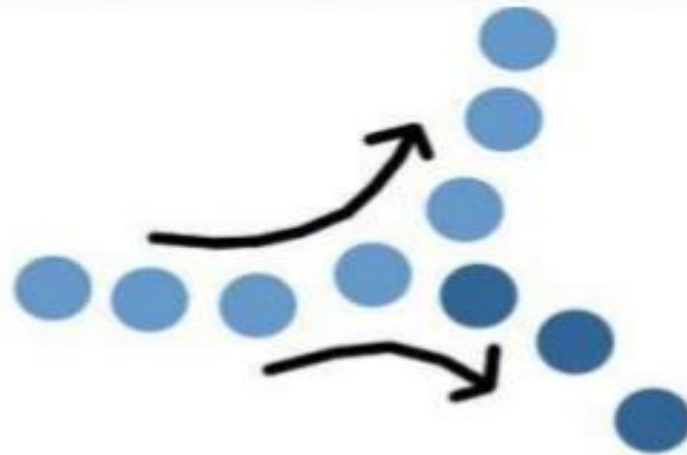
Law of Similarity —

The mind groups similar elements into collective entities or totalities. This similarity might depend on relationships of form, color, size, or brightness.

Chapter Two : laws of perception

Law of Good Continuation

Individuals have the tendency to continue contours whenever the elements of the pattern establish an implied direction. People tend to draw a good continuous line.



Law of Continuity:

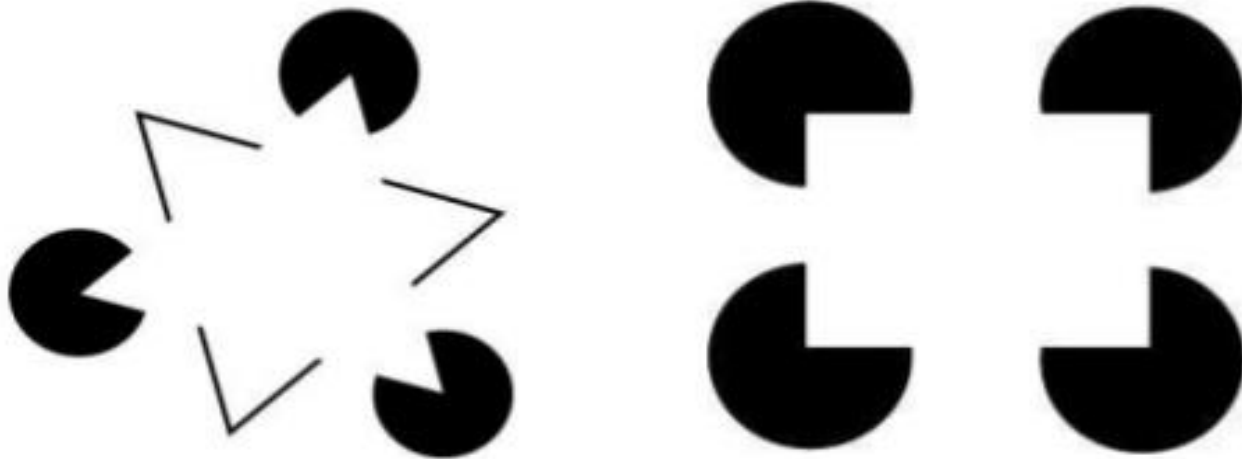
Lines are seen as following the smoothest path.

In the image above, the top branch is seen as continuing the first segment of the line. This allows us to see things as flowing smoothly without breaking lines up into multiple parts.

Chapter Two : laws of perception

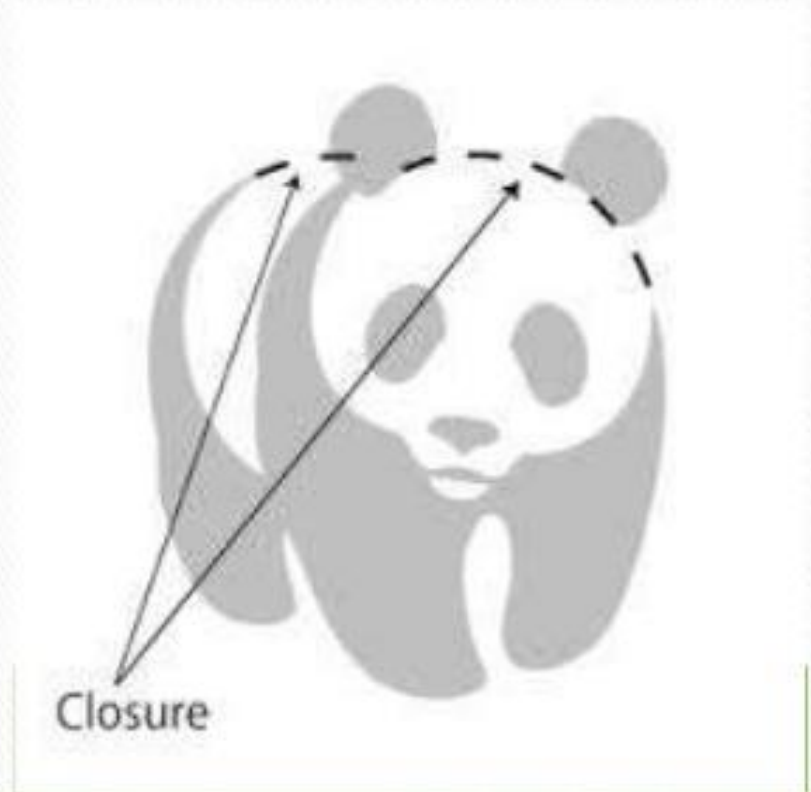
Law of Closure

- We tend to fill the gaps or “close” the figures we perceive. We enclose a space by completing a contour and ignoring gaps in the figure.



Chapter Two : laws of perception

Law of Closure



Our mind has the tendency to fill the gapes to get complete meaning

Chapter Two : laws of perception

Law of Closure



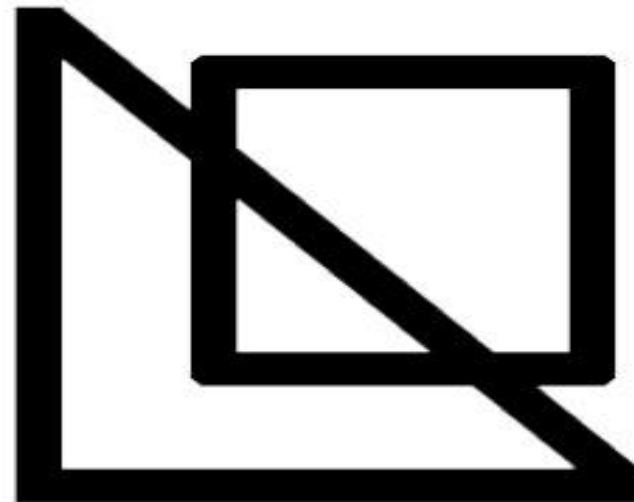
Design

Our mind has the tendency to fill the gapes to get complete meaning

Chapter Two : laws of perception

Law of Good Pragnanz

- ✓ The word *pragnanz* is a German term meaning "*good figure*." The law of Pragnanz is sometimes referred to as the law of good figure or the law of simplicity.
- ✓ This law holds that objects in the environment are seen in a way that makes them appear as simple as possible.



Chapter Two :Gestalt psychology and Perceptions

Law of Figure and Ground

Viewers will perceive an object (figure) and a surface (ground) even in shapes are grouped together.



Chapter Two :Gestalt psychology and Perceptions

Law of Figure and Ground

Viewers will perceive an object (figure) and a surface (ground) even in shapes are grouped together.



Chapter Two : perceptual constancy

Perceptual constancy refers to the tendency to perceive an object as remaining stable and unchanging ('constant') despite any changes that may occur to the image cast on the retina.

Chapter Two : perceptual constancy

Shape Constancy

Perceiving objects as unchanging even as illumination and retinal images change.
Perceptual constancies include constancies of shape and size.



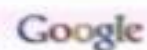
Shape Constancy



Chapter Two : perceptual constancy

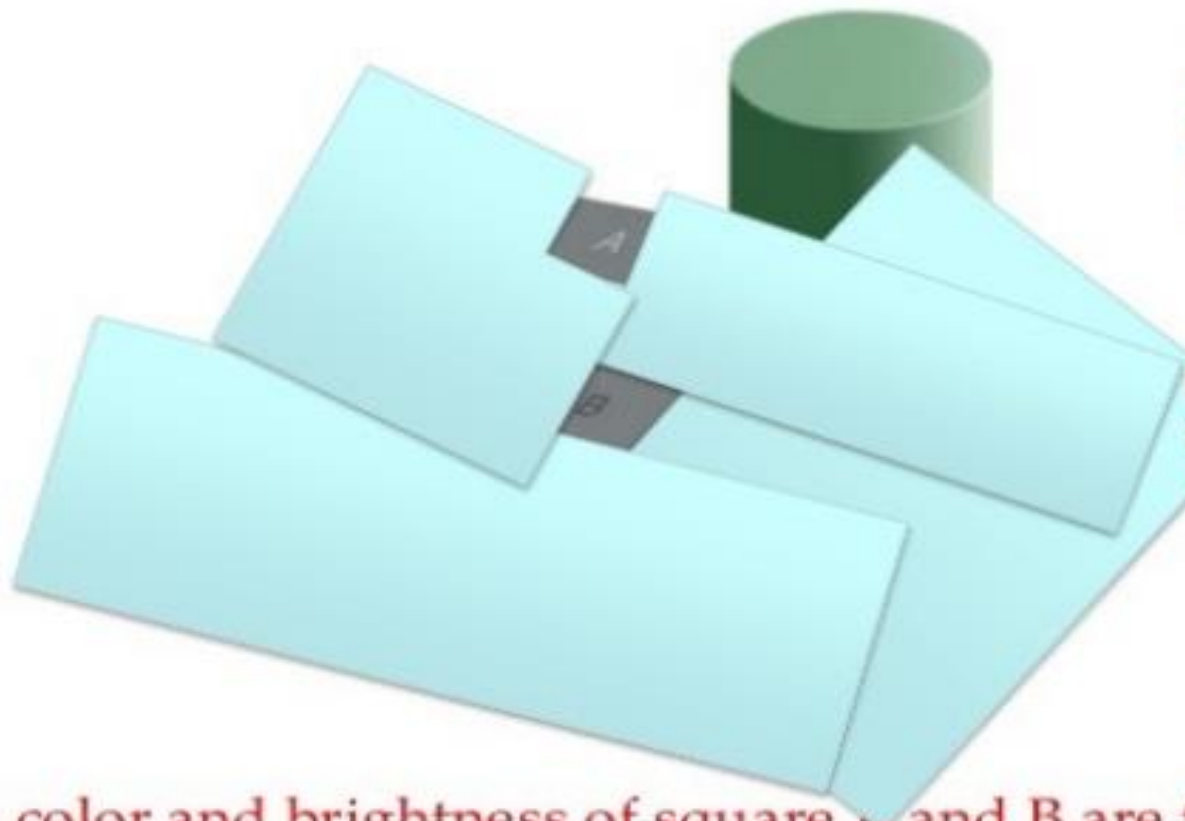
Size Constancy

Stable size perception amid changing size of the stimuli. We know the one car is just farther away, but still the same size.



Chapter Two : perceptual constancy

Lightness Constancy



Lightness constancy - enables us to perceive an object as having a constant lightness even when the light that falls on it changes.



The color and brightness of square A and B are the same.



Chapter Two: depth and Distance perception

Depth / Distance

▣ To observe/judge

Chapter Two: depth and Distance perception

CUES

- ▣ **Monocular Cues-** can only be seen with ONE eye.
- ▣ **Binocular Cues-** required to use BOTH eyes.

Depth perception depends on the use binocular cues and monocular cues there are two kinds of binocular cues: retinal disparity and convergence. The two kinds of binocular cues require the interaction of both eyes. Retinal disparity is, the degree of difference between the image of an object that are focused on the two retinas. The closer the object, the greater is the retinal disparity

To demonstrate retinal disparity for yourself, point a forefinger vertically between your eyes.

Chapter Two: depth and Distance perception

Monocular Cues

- ▣ Size cues
- ▣ The farther the smaller, the nearer the bigger.
- ▣ Retina-back part of the eye that receives the image of what is seen.

Chapter Two: depth and Distance perception

Linear Perspective

- ▣ Converging lines as it go farther
- ▣ Example: Railroad



Chapter Two: depth and Distance perception

Texture Gradient

- ▣ The farther the smoother



Chapter Two: depth and Distance perception

Atmospheric Perspective



Chapter Two: depth and Distance perception

Overlap(interposition)

- ▣ The ones at the back are smaller and farther, the ones in front the nearer and bigger.



Chapter Three : Learning

Learning

Definition of learning

Relatively permanent change in behavior as result of experiences
: training or practice

Chapter Three : Learning

The above definition emphasizes four attributes of learning:

- Learning is a change in behavior
- This change in behavior is relatively permanent
- It does not include change due to illness, fatigue, maturation and use of intoxicant.
- This permanent change in behavior is not because of biological factors (like hormonal
- changes) that bring permanent changes in behavior; but because of experience, or practice
- The learning is not directly observable but manifests in the activities of the individual.

Chapter Three : Learning

Characteristics of learning

1. Learning is continuous modification of behavior throughout life
2. Learning is pervasive, it reaches into all aspects of human life.
3. Learning involves the whole person, socially, emotionally & intellectually
4. Learning is often a change in the organization of experiences.
5. Learning is responsive to incentives
6. Learning is an active process
7. Learning is purposeful
8. Learning depends on maturation, motivation and practice.
9. Learning is multifaceted

Chapter Three : Learning

Principles of learning

- Individuals learn best when they are physically, mentally, and emotionally y ready to learn.
- Students learn best and retain information longer when they have meaningful practice and exercise
- Learning is strengthened when accompanied by a pleasant or satisfying feeling, and that
- learning is weakened when associated with an unpleasant feeling.
- Things learned first create a strong impression in the mind that is difficult to erase.

Chapter Three : Learning

...Principles of learning

- Things most recently learned are best remembered.
- The principle of intensity implies that a student will learn more from the real thing than from a substitute.
- Individuals must have some abilities and skills that may help them to learn.
- Things freely learned are best learned - the greater the freedom enjoyed by individuals, the
- higher the intellectual and moral advancement.

Chapter Three : Learning

Factors affecting learning

1. Motivation: When the motives of learning are high, the learner becomes enthusiastic.
2. Maturation: Neuro-muscular coordination is important for learning a given task.
3. Health condition of the learner: The learner should be in a good health status to learn.

Chapter Three : Learning

...Factors affecting learning

3. Psychological wellbeing of the learner: individual's psychological states like worries, fears, feelings of loneliness and inferiority hinders learning. Whereas self-respect, self-reliance, and self-confidence are necessary for effective learning.
4. Good working conditions – absence or presence of fresh air, light, comfortable surroundings, moderate temperature, absence of distractions like noise and learning aids determine learning effectiveness.

Chapter Three : Learning

Factors affecting learning

6. Background experiences: having background experiences affect effectiveness of learning. All related facts and understandings from a previously learned course should be brought to new learning.
7. Length of the working period: Learning periods should neither be too short nor too long. Long learning time sets fatigue and reduces effectiveness in learning. Short learning time doesn't allow adequate practice needed to master a learning task.
8. Massed and distributed learning: Learning that spreads across time with reasonable time gaps brings better results compared with crammed learning that occurs at once or within short span of time

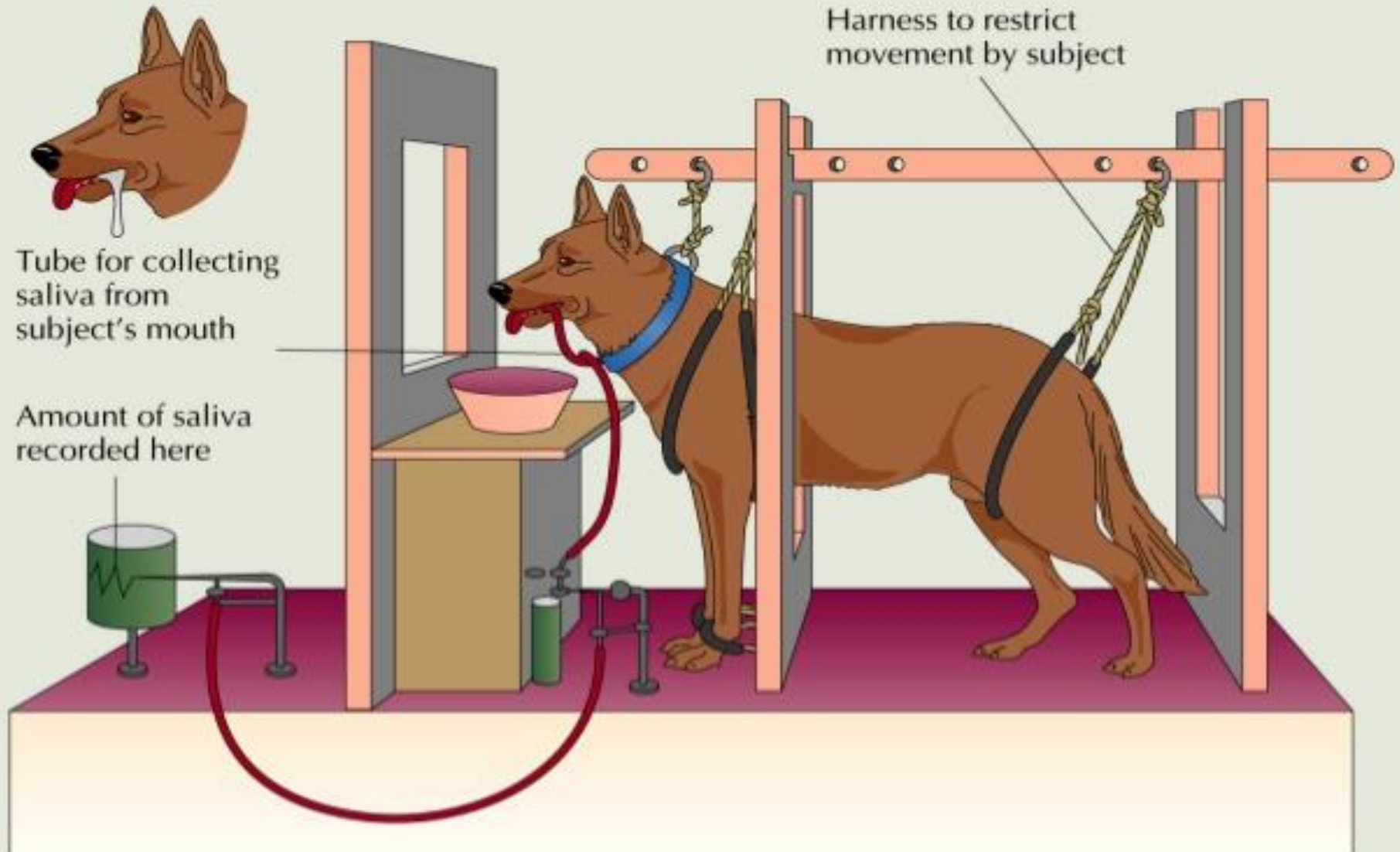
Chapter Three : Learning theories

1. Classical conditioning theory –behavioral
2. Operant conditioning – behavioral
3. Social Learning Theory (observational learning) theory
4. Cognitive learning may take two forms:
 1. Latent learning
 2. Insight learning (gestalt learning or perceptual learning)

Classical conditioning

- **Dogs normally respond to food by salivating .**
- **They do not have to be conditioned to salivate to food.**
- **Dogs do not , however , automatically salivate to the sound of a bell ringing .**
- **This is what Pavlov condition them to do.**
- **He re-ring the bell , present the food ,and the dogs would salivate .**
- **He repeated this procedure until the bell alone would cause the dogs to salivate .**
- **They had learned to associate the sound of the bell with the presentation food**

Classical conditioning; experiments



...Learning through association

Classical Conditioning

Simple definition: Learning to associate an automatic behavior (or feeling) with a stimulus.

Neutral Stimulus:

This stimulus does not naturally cause the subject to respond in a certain way.



Unconditioned Stimulus:

A stimulus that naturally (no learning involved) causes a particular response.



.....Basic elements

1- Classical Learning

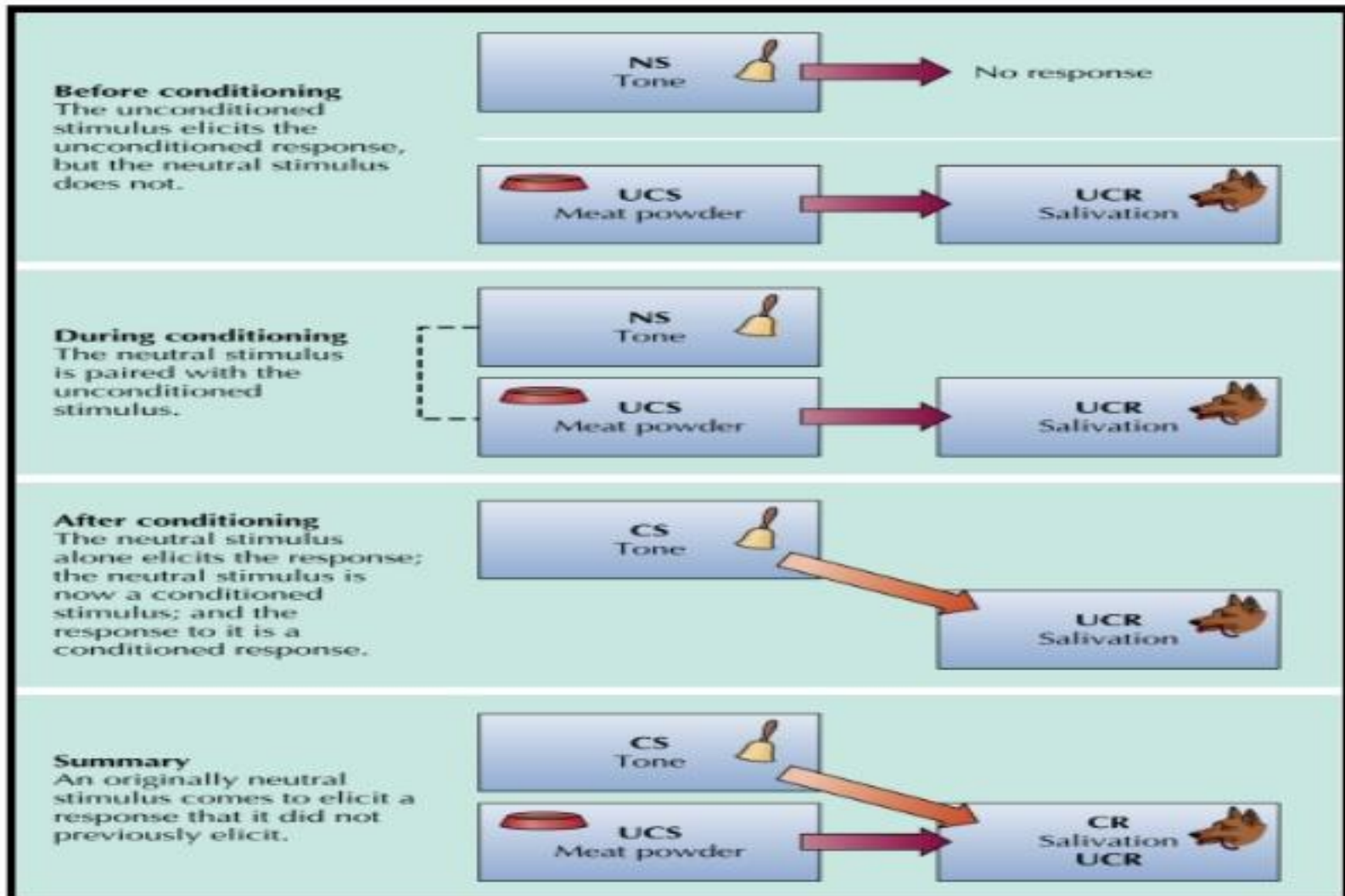
Unconditioned Stimulus (UCS) — the stimulus that automatically produce a reflex. **(in Pavlov's study this was the food).**

Unconditioned Response (UCR) — an automatic response to the UCS a natural response that does not require conditioning for it to occur **(in Pavlov's study this was salivation to the food).**

Conditioned Stimulus (CS) — a neutral stimulus that does not normally elicit an automatic response ; only after pairing it repeatedly with the UCS , does the CS come to elicit a conditioned response .
(in Pavlov's study this was the bell).

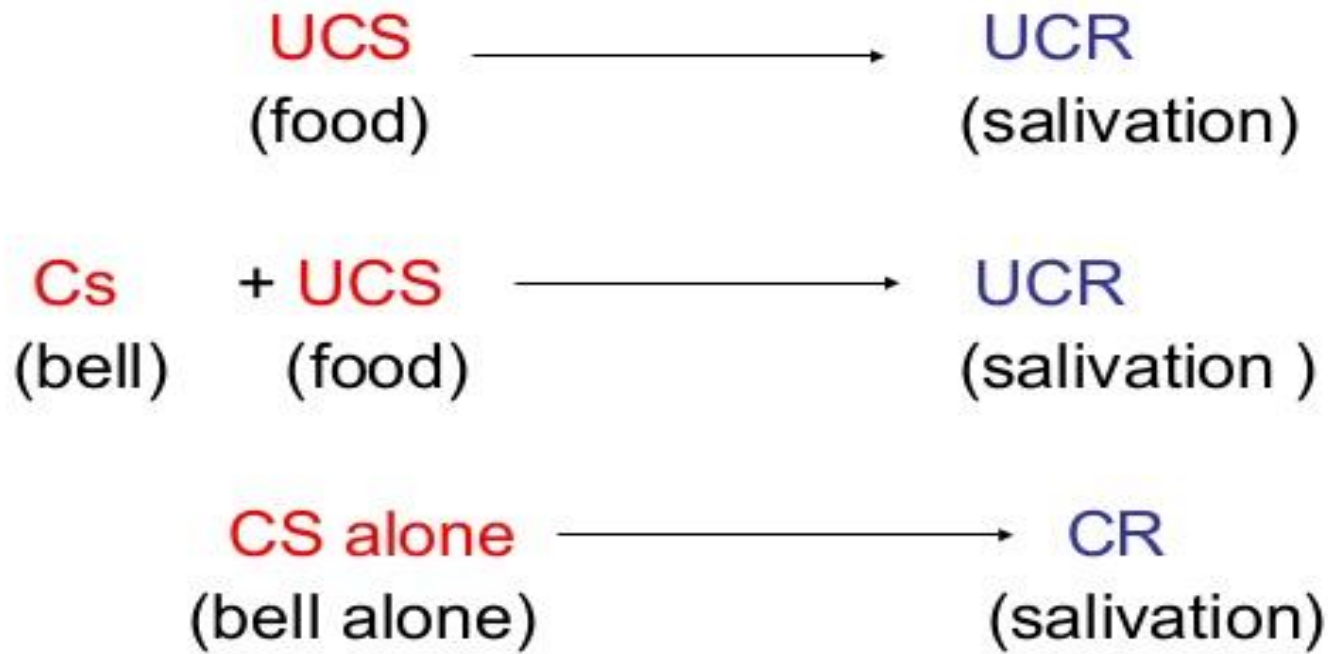
Conditioned Response (CR) — the learned response that occurs when the CS is presented alone , without the UCS .
(in Pavlov's study , the CR was salivation that occurred to the bell alone ; no food was present).

Stages of CC...



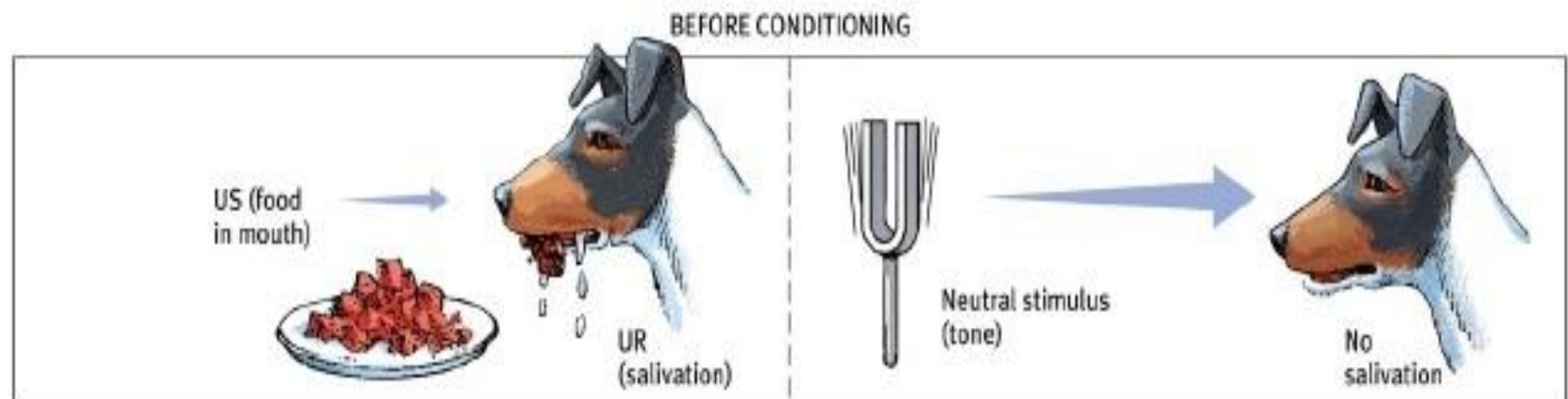
Stimulus –response association

Standard classical conditioning paradigm



Pavlov's Experiments

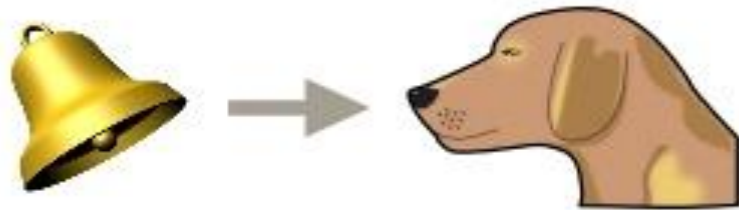
Before conditioning, food (Unconditioned Stimulus, US) produces salivation (Unconditioned Response, UR). However, the tone (neutral stimulus) does not.



An unconditioned stimulus (US) produces an unconditioned response (UR).

A neutral stimulus produces no salivation response.

1. Before Conditioning



Neutral
Stimulus:
Bell rings

No notable
response.

2. Before Conditioning



Unconditioned
Stimulus

Salivation
(Unconditioned
Response)

3. During Conditioning



Salivation
(Unconditioned
Response)

4. After Conditioning

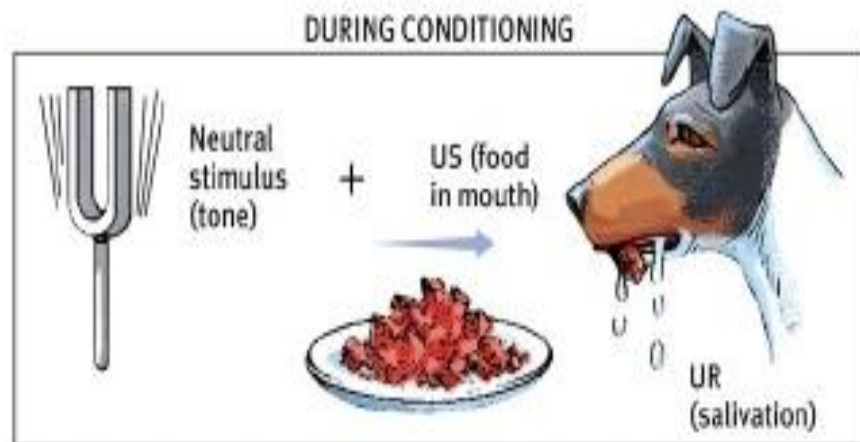


Conditioned
Stimulus

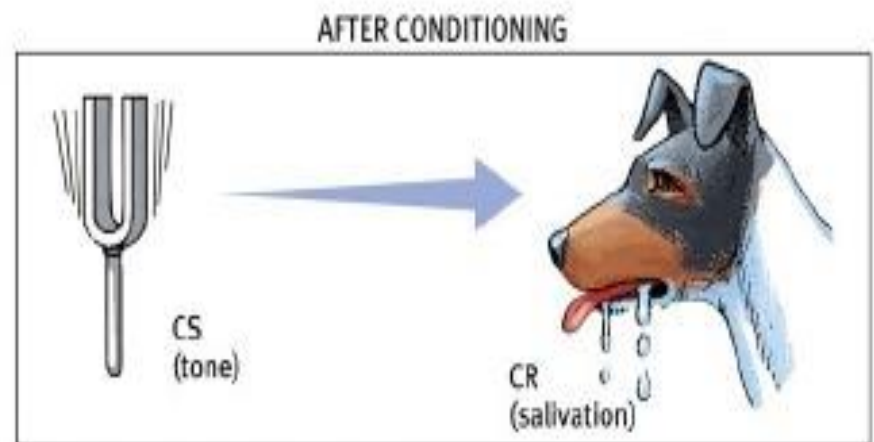
Salivation
(Conditioned
Response)

Pavlov's Experiments

During conditioning, the neutral stimulus (tone) and the US (food) are paired, resulting in salivation (UR). After conditioning, the neutral stimulus (now Conditioned Stimulus, CS) elicits salivation (now Conditioned Response, CR)



The unconditioned stimulus is repeatedly presented just after the neutral stimulus.



The neutral stimulus alone now produces a conditioned response

Chapter Three : Learning theories

- Principles of Classical Condition

The basic principles of classical conditioning include the role of stimulus generalization, stimulus discriminations, extinction and spontaneous recovery

A. Stimulus generalization and stimulus discrimination

Stimulus generalization is a process in which, after a stimulus has been conditioned to produce a particular response, stimuli that are similar to the original stimulus begin to produce the same responses. For example, a dog conditioned to salivate to a dinner bell (CS) might also salivate to a door bell, a telephone bell

Chapter Three : Learning theories

Principles of Classical Condition

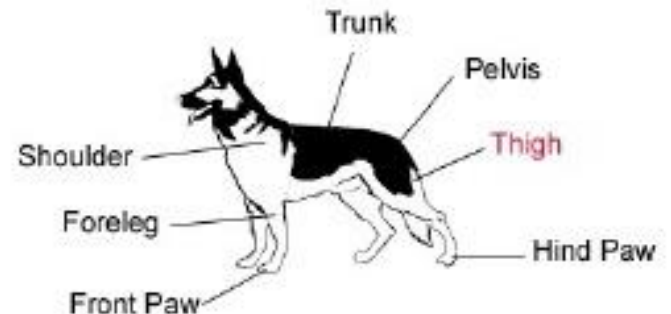
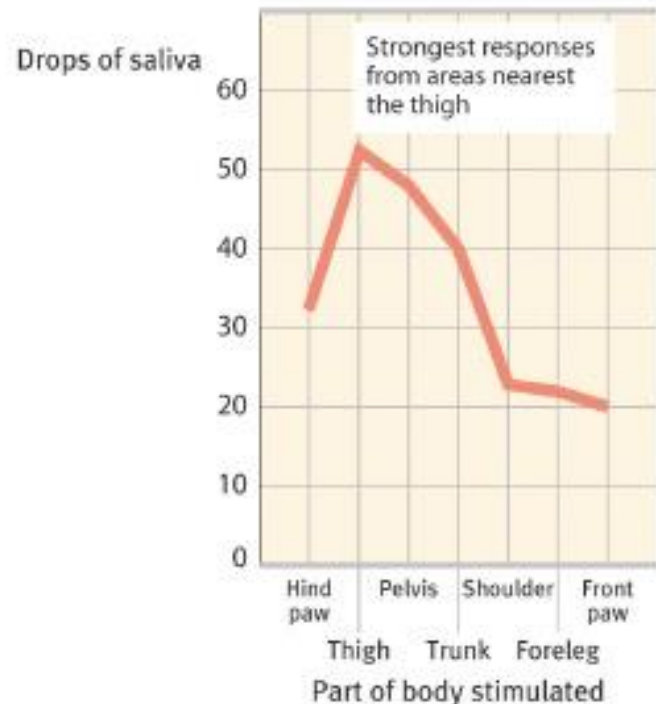
- **Stimulus discrimination-** is the process of distinguishing two similar stimuli; the ability to differentiate between stimuli.

Example, the dog salivates only in response to the dinner bell instead of the doorbell or the telephone bell

Stimulus Generalization

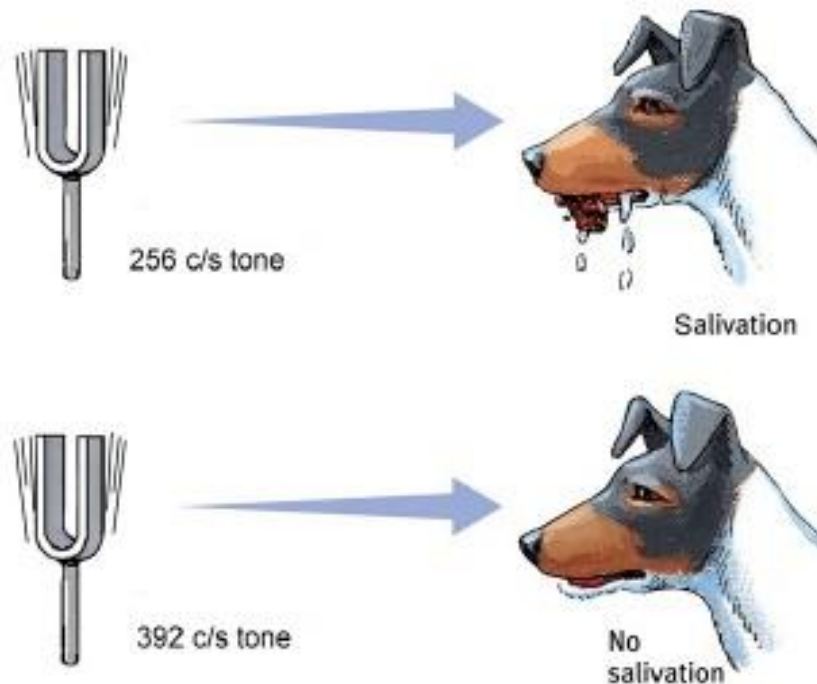
Tendency to respond to stimuli similar to the CS is called **generalization**.

Pavlov conditioned the dog's salivation (CR) by using miniature vibrators (CS) on the thigh. When he subsequently stimulated other parts of the dog's body, salivation dropped.



Stimulus Discrimination

Discrimination is the learned ability to distinguish between a conditioned stimulus and other stimuli that do not signal an unconditioned stimulus.



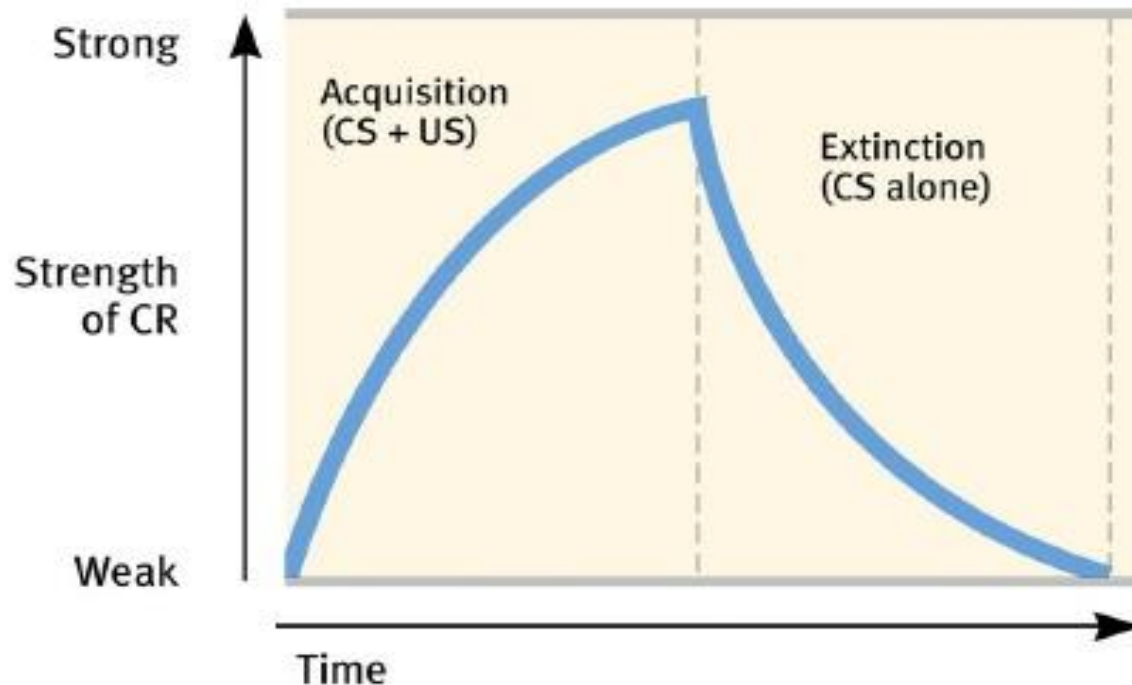
Chapter Three : Learning theories

Principles of Classical Condition

B. Extinction and spontaneous recovery

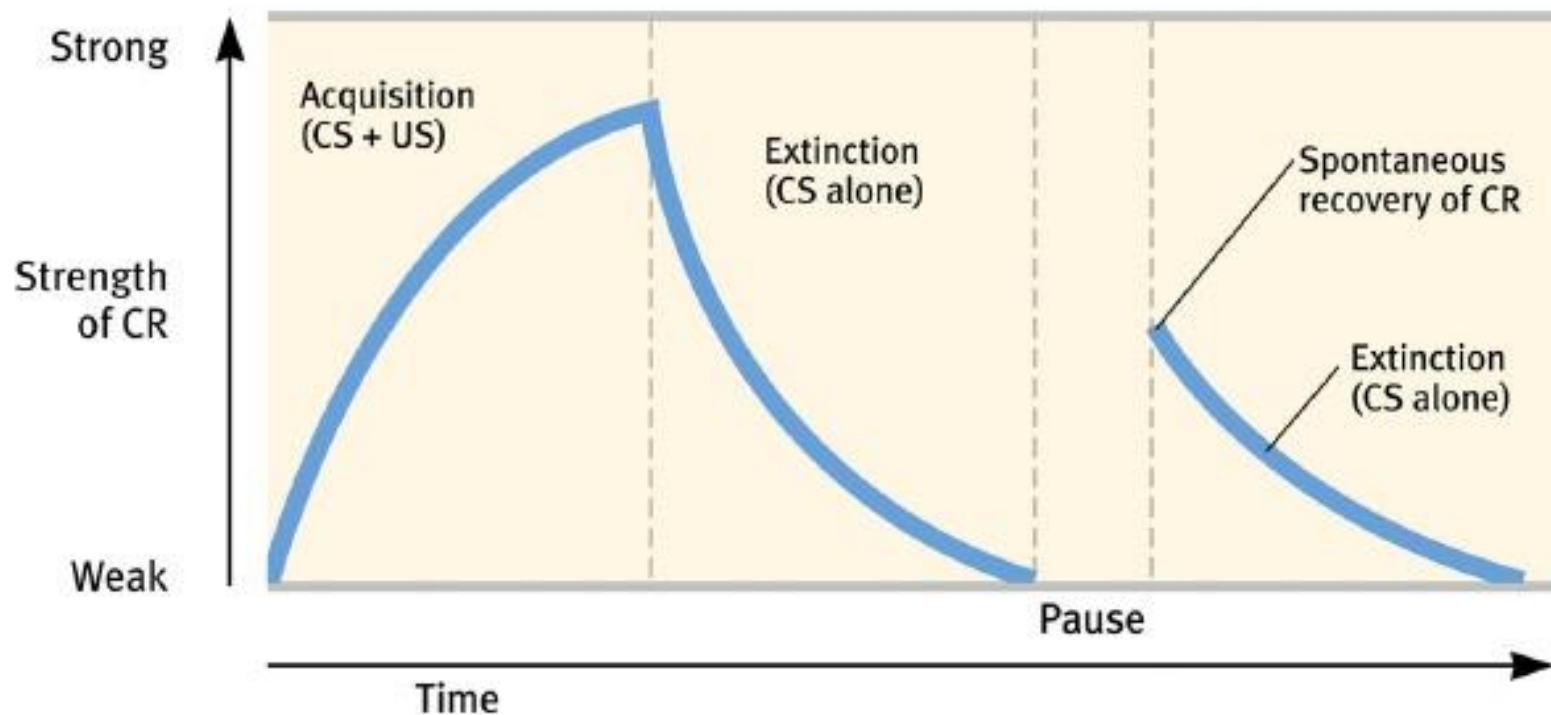
Extinction

When the US (food) does not follow the CS (tone), CR (salivation) begins to decrease and eventually causes extinction.



Spontaneous Recovery

After a rest period, an extinguished CR (salivation) spontaneously recovers, but if the CS (tone) persists alone, the CR becomes extinct again.



2. Operant conditioning

■ OPERANT

- Any active behavior that operates upon the environment to generate consequences



■ OPERANT CONDITIONING

- The behavior is followed by a consequence, and the nature of the consequence modifies the organisms tendency to repeat the behavior in the future

2. Operant conditioning

- In Skinner's analysis, a response (operant) can lead to three types of consequences: I. a neutral consequence II. a reinforcement and III. A punishment.
- A neutral Consequence that does not alter the response
- A reinforcement
- A Punishment

Operant Conditioning

A type of learning in which behavior is strengthened if followed by reinforcement or diminished if followed by punishment.



Cont...

Operant Conditioning

A method of learning that occurs through reinforcements and punishments for behavior. We learn to perform certain behaviors more often because they result in rewards, and learn to avoid other behaviors because they result in punishment or adverse consequences.



B.F. Skinner

expanded on Thorndike's work

shaping (reward approximations of the desired behavior)

VI. Burrhus Frederic (B. F.) Skinner

(March 20, 1904 – August 18, 1990)

- American psychologist, behaviorist, inventor, author and social philosopher.

- He was the Edgar Pierce Professor of Psychology at Harvard University from 1958 1974.



IV. Operant Conditioning



- is the type of learning in which behaviors are emitted to earn rewards or avoid punishment.

“Instrumental Learning”

2. Operant conditioning

- There are two basic types of reinforcers or reinforcing stimuli:
 - Primary reinforcers- Food, water, light, stroking of the skin, and a comfortable air temperature are naturally reinforcing because they satisfy biological needs.
 - Has the ability to strengthen a behavior without **prior learning**.
 - Secondary Reinforcers- Behaviors can be controlled by secondary reinforcers. They reinforce behavior because of their prior association with primary reinforcing stimuli.
 - Money, praise, applause, good grades, awards, and gold stars are common secondary reinforcers.

2. Operant conditioning

- Both primary and secondary reinforcers can be positive or negative.
- Positive reinforcement is the process whereby presentation of a stimulus makes behavior more likely to occur again.
- Negative reinforcement is the process whereby termination of an aversive stimulus makes behavior more likely to occur.
- The basic principle of negative reinforcement is that eliminating something aversive can itself be a reinforcer or a reward.
- Two types of negative reinforcement

Types of Negative reinforcement

▶ *Escape conditioning:*

Behavior is reinforced by the *elimination* of expected event from happening.

▶ *Avoidance conditioning:*

Behavior is reinforced by *preventing* an expected event from happening.

2. Operant conditioning

- **Schedules of reinforcement-** When a response is first acquired, learning is usually most rapid if the response is reinforced each time it occurs. This procedure is called **continuous reinforcement**. However, once a response has become reliable, it will be more resistant to extinction if it is rewarded on an intermittent (partial) schedule of reinforcement, which involves reinforcing only some responses, not all of them.
- **There are four types of intermittent schedules.**

2. Operant conditioning

1. **Fixed-ratio schedules-** occurs after a fixed number of responses. It produces high rate of responding.

- Employers often use fixed ratio schedules to increase productivity. An interesting feature of a fixed ratio schedule is that performance sometimes drops off just after reinforcement.

2. **Variable-Ratio Schedule-** occurs after some average number of responses but the number varies from reinforcement to reinforcement.

- A variable ratio schedule produces extremely high steady rates of responding. The responses are more resistant to extinction than a fixed ratio schedule is used.

2. Operant conditioning

3. **Fixed- Interval Schedule-** occurs only if a fixed amount of time has passed since the previous reinforcer.

4. **Variable- Interval Schedule-** occurs only if a variable amount of time has passed since the previous reinforcer.

- A basic principle of operant conditioning is that if you want a response to persist after it has been learned, you should reinforce it intermittently, not continuously

2. Operant conditioning

- **Punishment**- is a stimulus that weakens the response or makes it less likely to occur again. Punishers can be any aversive (unpleasant) stimuli that weaken responses or make them unlikely to reoccur . Like reinforcers, punishers can also be primary or secondary.
- Pain and extreme heat or cold are inherently punishing and are therefore known as primary punishers.
- Criticism, demerits, catcalls, scolding, fines, and bad grades are common secondary punishers.

2. Operant conditioning

- The positive-negative distinction can also be applied to punishment
- Something unpleasant may occur following some behavior (positive punishment), or
- something pleasant may be removed following the behavior (negative punishment).

2. Operant conditioning

The Pros and Cons of Punishment

- **Immediacy** – When punishment follows immediately after the behavior to be punished.
 - **Consistency**- when punishment is inconsistent the behavior being punished is intermittently reinforced and therefore becomes resistant to extinction.
 - **Intensity**- In general terms severe punishments are more effective than mild ones.
- ❑ However, there are studies that indicate that even less intense punishments are effective provided that they are applied immediately and consistently.

2. Operant conditioning

However, when punishment fails

1. People often administer punishment inappropriately or mindlessly
2. The recipient of punishment often responds with anxiety, fear or rage
3. The effectiveness of punishment is often temporary, depending heavily on the presence of the punishing person or circumstances
4. Most behavior is hard to punish immediately
5. Punishment conveys little information. An action intended to punish may instead be reinforcing because it brings attention

2. Operant conditioning

Shaping

- For a response to be reinforced, it must first occur. But, suppose you want to train a child to use a knife and a fork properly. Such behaviors, and most others in everyday life, have almost no probability of appearing spontaneously.
- The operant solution for this is shaping.
- **Shaping** is an operant conditioning procedure in which successive approximations of a desired response are reinforced. In shaping you start by reinforcing a tendency in the right direction. Then you gradually require responses that are more and more similar to the final desired response. The responses that you reinforce on the way to the final one are called successive approximations.

2. Operant conditioning

Applications of operant conditioning (reading assignment)

1. Conditioning study behavior
2. Conditioning and classroom behavior
3. Managing Problem Behavior
4. Dealing with anxieties through conditioning
5. Conditioning group behavior
6. Conditioning and Cognitive Processes
7. Shaping Complex Behavior

Chapter 3: learning

Social Learning Theory (observational learning) theory

- According to Bandura, a major part of human learning consists of observational learning, which is learning by watching the behavior of another person, or the model.
- Because of its reliance on observation of others, a social phenomenon; the perspective taken by Bandura is often referred to as a social cognitive approach to learning (Bandura, 1999, 2004).

ALBERT BANDURA



- Born on December 4, 1925
- Canadian, American
- Studied at University of British Columbia and University of Iowa.
- Researched and taught at Stanford University
- Elected president of American Psychological Association in 1974
- Famous for research on social learning theories.

“..most learning is gained by people’s perception and thinking about what they experience. They learn by copying the examples of others around them.”

- Albert Bandura



Chapter 3: learning

- Bandura identifies three forms of reinforcement that can encourage observational learning.
- direct reinforcement
- vicarious reinforcement – not direct or indirect
- self-reinforcement, or controlling

Chapter 3: learning

4 conditions of the of the modeling process

1. Attention
2. Retention- remembering the observed behavior
3. Motor reproduction- demonstrated
4. Motivation – approval

Chapter 3: learning

Classroom implications (reading assignment)

1. Students often learn a great deal simply by observing other people
2. Describing the consequences of behavior can effectively increase the appropriate behaviors and decrease inappropriate ones
3. Modeling provides an alternative to shaping for teaching new behaviors

Chapter 3: learning

...Classroom implications (reading assignment)

4. Teachers and parents must model appropriate behaviors and take care that they do not model inappropriate behaviors.
5. Teachers should expose students to a variety of other models
6. Students must believe that they are capable of accomplishing school tasks.

Chapter 3: learning

... Classroom implications (reading assignment)

7. Teachers should help students set realistic expectations for their academic accomplishments

Chapter 3: learning

Cognitive Learning Theory

1. Latent Learning

- Latent means hidden and thus latent learning is learning that occurs but is not evident in behavior until later, when conditions for its appearance are favorable. It is said to occur without reinforcement of particular responses and seems to involve changes in the way information is processed

2. Insight Learning

- It is a cognitive process whereby we reorganize our perception of a problem. It doesn't depend on conditioning of particular behaviors for its occurrence. Sometimes, for example, people even wake up from sleep with a solution to a problem that they had not been able to solve during the day.

CHAPTER FOUR MEMORY AND FORGETTING